

Pearson Edexcel

Level 3 GCE

Practice Paper (Hard Level)

(Time: 1 hour 30 minutes)

Paper Reference: 8EB0/02 (Practice)

Economics B

Advanced Subsidiary

Paper 2: The wider economic environment

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided - *there may be more space than you need.*

Information

- The total mark for this paper is **80**.
- The marks for **each** question are shown in brackets - *use this as a guide as to how much time to spend on each question.*
- Calculators may be used.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Answer ALL questions.

SECTION A

Read the following extracts (A and B) before answering Question 1.

Write your answers in the spaces provided.

Extract A

UK car industry faces challenging transition to electric vehicles

The UK automotive industry is undergoing a profound structural transformation. Government policy now requires that all new cars sold in the UK must be zero emission by 2035, forcing manufacturers to invest heavily in electric vehicle (EV) production. This transition is occurring against a backdrop of weak macroeconomic conditions and rising costs.

Many established UK car manufacturers, including Jaguar Land Rover and Mini, have committed billions of pounds to EV production lines. However, the cost of producing electric vehicles remains significantly higher than for conventional petrol or diesel vehicles, partly due to the high cost of lithium-ion batteries, which can account for up to 40% of an EV's total production cost.

UK consumers have been cautious about adopting electric vehicles. Despite government subsidies and tax incentives, the average price of a new EV in the UK is around £45,000, compared with £29,000 for a petrol equivalent. The growth in real disposable income across UK households has been weak over the last decade, making large purchases such as new cars more difficult for many families.

The annual percentage change in average nominal earnings was 6.45% in October 2023. UK inflation in the same month was 4.62%. While real wages have started to rise, consumer confidence remains fragile, and household savings rates have fallen as families have used reserves to cope with higher living costs.

UK car manufacturers also face higher operating costs. Energy prices remain elevated, and post-Brexit trade arrangements have added complexity to supply chains. EU workforce shortages have driven up wages for skilled engineering staff, while the National Living Wage continues to rise, affecting employment costs across automotive supply chains.

Mike Hawes, Chief Executive of the Society of Motor Manufacturers and Traders, said: "The transition to electric vehicles represents the biggest structural change the UK automotive sector has faced in a century. While the long-term direction of travel is clear, aggregate demand for new vehicles remains subdued and manufacturers must navigate significant short-term pressures."

(Source: adapted from industry reports, 2023)

Extract B

UK labour market statistics: October 2023

Labour force	34,278,000
Employed	32,801,000

(Source: adapted from Office for National Statistics, 2023)

1 (a) What is meant by the term 'structural change'? **(2)**

(b) Using Extract B, calculate, to 2 decimal places, the rate of unemployment for the UK in October 2023. **(4)**

(c) Using Extract A, calculate, to 2 decimal places, the change in real earnings for October 2023. **(4)**

(d) Discuss the possible benefits for consumers of the transition to electric vehicles. **(8)**

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This image shows a full page of blank, lined paper. It features approximately 20 evenly spaced horizontal grey lines across its entire width, providing a guide for handwriting or typing. The paper itself is a clean, off-white color.

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(Total for Question 1 = 30 marks)

TOTAL FOR SECTION A = 30 MARKS

SECTION B

Read the following extracts (C, D and E) before answering Question 2.

Write your answers in the spaces provided.

Extract C

The rapid growth of Octopus Energy in the UK energy market

Octopus Energy, founded in 2015, has emerged as one of the fastest-growing energy suppliers in the UK. The company now serves more than 6.5 million UK households and has expanded internationally into Germany, France, the United States and Japan. Octopus differentiates itself from traditional energy companies through its strong focus on renewable energy and its investment in smart digital technology.

Octopus has built its reputation on customer service and on offering tariffs that reward consumers for using electricity at off-peak times. Its 'Agile' tariff allows customers to pay variable prices that change every 30 minutes, reflecting wholesale energy costs. This has been particularly attractive to households with electric vehicles, who can charge cars overnight when prices are low.

In 2022, Octopus acquired Bulb Energy, taking on around 1.5 million customers and significantly expanding its market share. The company now competes directly with British Gas and E.ON in what was previously a market dominated by long-established providers. Many industry observers initially regarded Octopus as a niche operator targeting environmentally conscious consumers, but the company has now moved firmly into the mass market.

A standard Octopus electricity tariff costs approximately 27p per kWh, broadly in line with the energy price cap set by Ofgem. However, customers on the Agile tariff have reported saving up to 40% on their electricity bills compared with standard tariffs.

The original Octopus product, the standard variable tariff, is widely regarded as having reached the maturity stage of its product life cycle. The company has therefore prioritised expansion into new categories, including heat pumps, solar panel installation and battery storage systems.

(Source: adapted from industry analysis, 2023)

Extract D

Octopus invests in new technology to support growth

Octopus has invested over £1 billion in Kraken, its proprietary energy management software platform. Kraken is now licensed to other energy suppliers across the world, generating an additional revenue stream for the company.

However, rapid expansion has placed significant pressure on the company's operations. Customer service teams have reportedly been overwhelmed by the integration of former Bulb customers, with call waiting times rising sharply during peak periods. Octopus has acknowledged that some aspects of its business are now operating at over-utilisation of capacity.

The company is recruiting an additional 800 customer service staff to cope with demand, but training new employees takes time. In the meantime, customer satisfaction scores have shown a noticeable decline.

A senior Octopus executive said: "Growing rapidly while maintaining service quality is one of the toughest challenges in business. We are confident we can manage the transition, but we have to be honest that the short-term pressures are significant."

(Source: adapted from industry analysis, 2023)

Extract E

Octopus launches new heat pump service with introductory pricing

Octopus is launching a new heat pump installation service across the UK, with introductory pricing designed to encourage rapid adoption. Customers signing up in the first six months will pay £2,500 for a complete installation, compared with the standard market price of approximately £7,000.

Octopus has been able to offer this aggressive pricing due to government subsidies under the Boiler Upgrade Scheme, combined with the company's own investment in installer training and supply chain efficiency.

The marketing director said: "We believe heat pumps will play a critical role in the UK's transition to net zero. By using penetration pricing, we hope to make the technology accessible to a much wider range of customers and to establish Octopus as the leading installer in the UK market."

(Source: adapted from press release, 2023)

2 (a) What is meant by the term 'mass market'? **(2)**

Octopus has reduced the price of one of its electricity tariffs by 12%. This led to a 9% increase in customer numbers.

(b) Calculate the price elasticity of demand (PED) for this tariff. You are advised to show your working. **(4)**

(c) Explain one possible benefit for Octopus of investing heavily in its Kraken software platform. **(4)**

(d) Discuss the implications of over-utilisation of capacity for Octopus. **(8)**

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(Total for Question 2 = 30 marks)

TOTAL FOR SECTION B = 30 MARKS

SECTION C

Read the following extract (F) before answering Question 3.

Write your answer in the space provided.

Extract F

The development dilemma: balancing economic growth and environmental protection in emerging economies

Many emerging economies, including India, Vietnam and Indonesia, are pursuing rapid economic development to lift hundreds of millions of citizens out of poverty. However, this rapid growth has come at a significant environmental cost. Air pollution levels in major cities such as New Delhi and Jakarta now regularly exceed World Health Organization guidelines by more than ten times, contributing to an estimated 7 million premature deaths globally each year.

India's economy has grown at an average rate of around 6% per year over the past decade, with much of this growth driven by manufacturing and construction. Coal continues to provide more than 70% of India's electricity, and emissions of greenhouse gases have risen sharply. The Indian government argues that developed countries, which industrialised using fossil fuels, should not now expect developing nations to forgo the same path.

Environmental pressure groups argue that there is a clear trade-off between economic growth and negative externalities. Industrial production generates pollution that imposes external costs on third parties, including respiratory illness, contaminated water supplies and damage to ecosystems. The economic and social costs of these externalities can be substantial: the World Bank estimates that the cost of environmental degradation in India is equivalent to around 5.7% of GDP per year.

However, supporters of continued economic development argue that growth is essential to allow developing economies to invest in cleaner technologies. As countries grow richer, they typically reach a point at which they can afford to introduce stricter environmental regulations and invest in renewable energy. This relationship, known as the Environmental Kuznets Curve, suggests that pollution often rises in the early stages of development before falling as countries become wealthier.

Government policies in emerging economies have begun to address these concerns. India has set a target of reaching net zero greenhouse gas emissions by 2070 and has committed to installing 500 GW of renewable energy capacity by 2030. Indonesia is investing heavily in electric vehicle production and battery manufacturing. Vietnam has committed to phasing out new coal-fired power stations.

Critics argue that these commitments may not be enough. The scale of investment required is enormous, and the political pressure to maintain rapid economic growth remains strong. There is also concern that wealthy countries have not done enough to support emerging economies through technology transfer and climate finance, leaving developing countries to bear the costs of transition.

(Source: adapted from international development reports, 2023)

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(Total for Question 3 = 20 marks)

TOTAL FOR SECTION C = 20 MARKS

TOTAL FOR PAPER = 80 MARKS